Queue Using Linked List

Program:

```
#include<stdio.h>
#include<stdlib.h>

/* Structure to create a node with data and next pointer */

struct node
 {
   int data;
   struct node *next;
 };

struct node *front = NULL;
struct node *rear = NULL;

/* Enqueue operation on a queue */

void enqueue(int value)
{

struct node *ptr;
ptr = (struct node*)malloc(sizeof(struct node));
ptr->data = value;
ptr->next = NULL;

if((front == NULL) && (rear == NULL)) {
front rear = ptr;
}
else
{
rear->next = ptr;
rear = ptr;
}
}
printf("Node is Inserted:%d\n\n",value);

/* Dequeue operation on a queue */

int dequeue()
{
if(front == NULL)
```

```
{ printf("\nUnderflow\n");
return -1;
}
else
{
struct node *temp = front;
int temp_data = front->data;
front = front->next;
free(temp);
return temp_data;

/* Display all elements of queue */
void display()
{
struct node *temp;
if((front == NULL) && (rear == NULL)) {
printf("\nQueue is Empty\n");
}
else
}

{
printf("The queue is \n");
temp = front;
while(temp)
printf("%d--->",temp->data);
temp = temp->next;
}
printf("NULL\n\n");
}
}

int main()
{
enqueue(10);
enqueue(20);
enqueue(30);
display();
}
dequeue();
display(); return 0;
```

Output:
Node is Inserted:10

Node is Inserted:20
Node is Inserted:30
The queue is 10--->20--->30--->NULL
0.72
The queue is 20--->30--->NULL